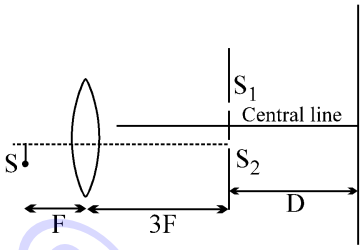
SECTION-I

Single Correct Answer Type

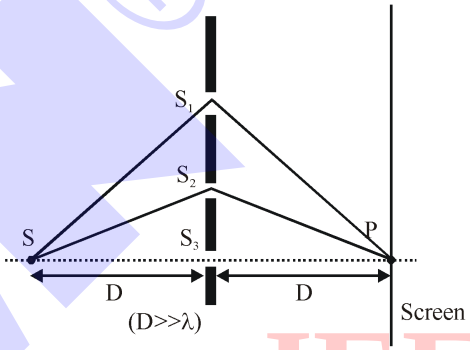
8 Q. [3 M (–1)]

1. Observe the set up shown in the figure. What will be the position of the central maxima on the screen? Source S is d/2 distance below the optical axis and the optical axis is equal distance below the central line. (The separation between the slits is d and separation between the plane of slit and the screen is D and focal length of lens is F.) Also F >> d, D >>d



(A) $\frac{Dd}{2F}$ (B) $\frac{Dd}{F}$ (C) $\frac{Dd}{3F}$ (D) None

2. A monochromatic light source of wavelength λ is placed at S. Three slits S_1, S_2 and S_3, source S and point P is shown in the figure. If $S_1P - S_2P = \lambda/6$ and $S_1P - S_3P = 2\lambda/3$ and I be the intensity at P when only one slit is open, the intensity at P when all the three slits are open is :-

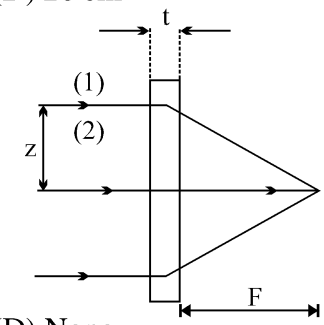


(A) 3I
(B) 5I
(C) 8I
(D) 0

3. A parallel beam of light is incident on an equi-convex lens having radius of curvature R = 20 cm. Find the radius of curvature of emergent wavefront, just after emerging from lens: (reflective index of lens = 1.5)

(A) 10 cm (B) 5 cm (C) 1 cm (D) 20 cm

4. Plane wavefronts are incident on a glass slab which has refractive index as a function of distance Z, according to the relation $\mu = \mu_0^- (1 - Z^2/Z_0^2)$, where μ_0 is the refractive index along the axis and Z_0 is a constant. This glass slab can act as lens of focal length F. By using the concept of optical path length calculate the focal length of the slab. Consider t to be very small as compared to F.
[**Hint :** Equate the OPL of ray (1) and ray (2)]

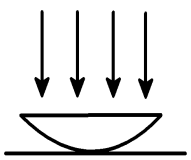


(A) $Z_0^2/(2\mu_0 t)$ (B) $Z_0^2/(\mu_0 t)$ (C) $\mu_0 Z_0^2/(2t)$ (D) None

5. Monochromatic light of wavelength 900 nm is used in a young's double slit experiment. One of the slits is covered by a transparent sheet of thickness 1.8×10^{-5} m made of material of refractive index 1.6. How many fringes will shift due to introduction of the sheet :-

(A) 18 (B) 12 (C) 10 (D) 6

6. A thin slice is cut out of a glass cylinder along a plane parallel to its axis. The slice is placed on a flat glass plate as shown. The observed interference fringes from this combination shall be



(A) straight

(B) circular

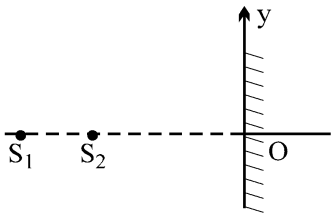
(C) equally spaced

(D) having fringe spacing which increases as we go outwards.

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7. Two point monochromatic and coherent sources of light of wavelength λ are placed on the dotted line in front of a large screen. The sources emit waves in phase with each other. The distance between S_1 and S_2 is d while their distance from the screen is much larger.

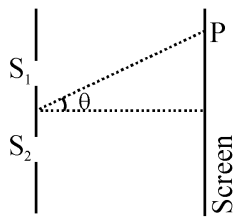


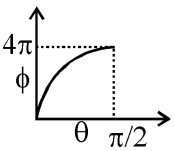
(1) If $d = 7\lambda/2$, O will be a minima
(2) If $d = 4.3\lambda$, there will be a total of 8 minima on y-axis.
(3) If $d = 7\lambda$, O will be a maxima.
(4) If $d = \lambda$, there will be only one maxima on the screen.
Which is the set of correct statement :
(A) 1, 2 & 3 (B) 2, 3 & 4 (C) 1, 2, 3 & 4 (D) 1, 3 & 4

8. Direction :
The question has a paragraph followed by two statement, Statement-1 and statement-2. Of the given four alternatives after the statements, choose the one that describes the statements.
A thin air film is formed by putting the convex surface of a plane-convex lens over a plane glass plate. With monochromatic light, this film gives an interference pattern due to light reflected from the top (convex) surface and the bottom (glass plate) surface of the film
Statement-1: When light reflects from the air-glass plate interface, the reflected wave suffers a phase change of π.
Statement-2 : The centre of the interference pattern is dark :-
(A) Statement-1 is true, Statement-2 is true and Statement-2 is not the correct explanation of Statement-1.
(B) Statement-1 is false, Statement-2 is true
(C) Statement-1 is true, Statement-2 is false
(D) Statement-1 is true, Statement-2 is true and Statement-2 is the correct explanation of statement-1.

Multiple Correct Answer Type 8 Q. [4 M (–1)]

9. In a YDSE setup, we plot the phase difference (ϕ) between both waves at point P on the screen against the angular position (θ) of point P on the screen. The graph is as shown below.

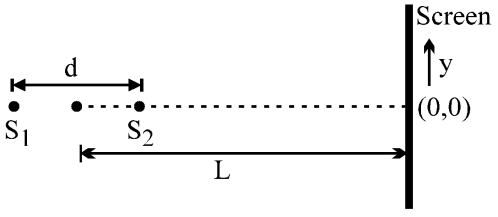




(A) The distance $S_1S_2 = 2\lambda$
(B) There are a total of 4 minima on the screen
(C) The first maxima above the centre is at $\theta = \frac{\pi}{4}$
(D) At $\theta = \frac{\pi}{6}$, intensity is maximum

10. In Young's Double slit experiment, the interference pattern is found to have an intensity ratio between the bright and dark fringe is 9. This implies that
(A)the intensities at the screen due to the two slits are 5 and 4 units respectively.
(B) the intensities at the screen due to the two slits are 4 and 1 units respectively.
(C) the amplitude ratio is 3
(D) the amplitude ratio is 2

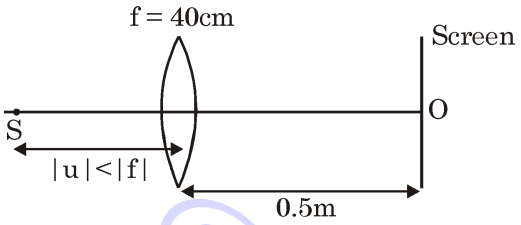
11. The figure shows two points source which emit light of wavelength λ in phase with each other and are at a distance $d = 5.5\lambda$ apart along a line which is perpendicular to a large screen at a distance L from the centre of the source. Assume that d is much less than L. Which of the following statement is (are) correct ?



(A) Five bright fringes appear on the screen
(B) Six bright fringes appear on the screen
(C) Point y = 0 corresponds to bright fringe
(D) Point y = 0 corresponds to dark fringe.

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12. A lens of focal length f = 40 cm is cut along the diameter into two equal halves. In this process, a layer of thickness t = 1mm is lost, then halves are put together to form a composite lens. In between focal plane and the composite lens a narrow slit is placed very close to the focal plane |u| < |f|. The slit is emitting monochromatic light of wavelength 0.6 μm. Behind the lens a screen is located at a distance L = 0.5 m from it as shown :-



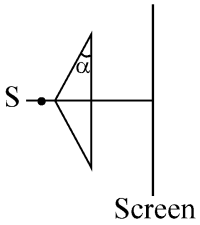
(A) Fringe width is 0.12 mm

(B) Fringe width is 0.24 mm.

(C) Length of interference pattern is 1/8 cm

(D) Length of interference pattern is 1/16 cm

13. For the biprism experiment shown in the figure, the fringe width increases when



(A) biprism is moved towards the slit

(B) entire apparatus is submerged in a liquid having R.I. less than that of prism

(C) a biprism having smaller angle α is used

(D) the slit width is reduced

14. When monochromatic light is incident normally on a wedge-shaped thin air film, refer figure, an interference pattern may be seen by reflection. Which of the following is/are correct?



(A) Parallel fringes are observed

(B) If water is introduced into the region between the plates, the fringe separation decreases

(C) If the angle of the wedge is increased, the fringe separation decreases

(D) When white light is used there will not be a completely dark fringe

15. In an arrangement to view the interference pattern produced by a wedge of liquid between a microscope slide and a glass block the interference pattern obtained is made of equally spaced parallel fringes. The fringe separation (or fringe width) can not be decreased by which of the following actions.

(A) by increasing the angle of the liquid wedge.

(B) by using a liquid of smaller refractive index.

(C) by using a thicker glass block.

(D) by using a longer liquid wedge of the same angle

16. In a Young's double slit experiment, the separation between the two slits is d and the wavelength of the light is λ. The intensity of light falling on slit 1 is four times the intensity of light falling on slit 2. Choose the correct choice(s).

(A) If $d = \lambda$, the screen will contain only one maximum

(B) If $\lambda < d < 2\lambda$, at least one more maximum (besides the central max.) will be observed on the screen

(C) If the intensity of light falling on slit 1 is reduced so that it becomes equal to that of slit 2, the intensities of the observed dark and bright fringes will increase

(D) If the intensity of light falling on slit 2 is increased so that it becomes equal to that of slit 1, the intensities of the observed dark and bright fringes will increase

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SECTION-II

Numerical Answer Type Question **3Q.[3M(0)]**

(upto second decimal place)

17. In a Young's experiment, the upper slit is covered by a thin glass plate of refractive index 1.4 while the lower slit is covered by another glass plate having the same thickness as the first one but having refractive index 1.7 . Interference pattern is observed using light of wavelength 5400 Å . It is found that the point P on the screen where the central maximum (n = 0) fell before the glass plates were inserted now has 3/4 the original intensity . It is further observed that what used to be the 5th maximum earlier, lies below the point P while the 6th minimum lies above P. Calculate the thickness of the glass plate.
(Absorption of light by glass plate may be neglected).

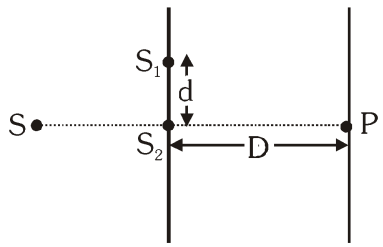
18. A thin film of a specific material can be used to decrease the intensity of reflected light. There is destructive interference of waves reflected from upper and lower surfaces of the film. These films are called anti-reflection coating. Magnesium fluoride (MgF_2) is used as anti-reflecting coating on plane glass surface for light having wavelength 500 nm and normal incidence. The minimum thickness of film required is 100 nm. Find refractive index of magnesium fluoride. It is known that refractive index of glass is more than refractive index of magnesium fluoride.

19. In a YDSE apparatus, $d = 1$mm, $\lambda = 600$ nm and $D = 1$m. The slits individually produce same intensity on the screen. Find the minimum distance between two points on the screen having 75% intensity of the maximum intensity.

SECTION-III

Numerical Grid Type (Ranging from 0 to 9) **1 Q. [4 M (0)]**

20. In a YDSE experiment two slits S_1 and S_2 have separation of d = 2 mm. The distance of the screen is D = 8/5 m. Source S starts moving from a very large distance towards S_2 perpendicular to S_1S_2 as shown in figure. The wavelength of monochromatic light is 500 nm. The number of maximas observed on the screen at point P as the source moves towards S_2 is 3995 + n. Find the value of n.

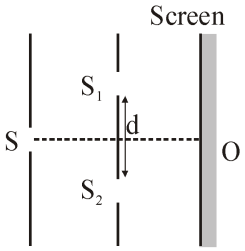
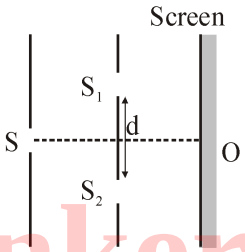
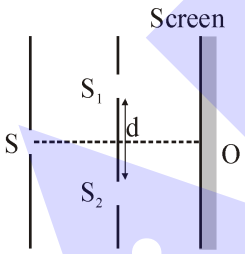
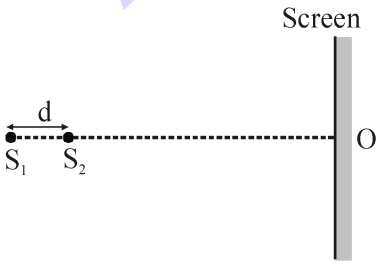


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SECTION-IV

Matrix Match Type (4 × 5) **3 Q. [8 M (for each entry +2(0)]**

21. Column-I describes various arrangements to obtain interference pattern on screen.

	Column I	**Column II**
(A)	 S_1, S_2 are pin holes. d = 100 λ	(P) Centre of screen 'O' is dark.
(B)	 S_1, S_2 are slits. d = 100.5 λ	(Q) First order maxima will not be seen.
(C)	 S_1, S_2 are pin holes. d = 0.5 λ	(R) Shape of fringes is hyperbolic
(D)	 S_1, S_2 are coherent point sources with zero initial phase difference. d = 2.5 λ	(S) Shape of fringes is circular. (T) Shape of fringes is linear.

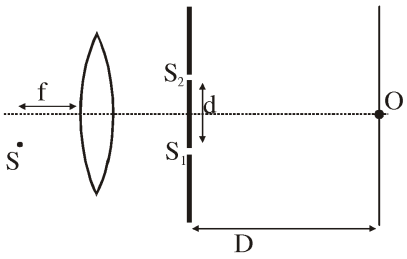
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22. Column-I shows some modifications in a standard YDSE setup. Column-II shows the associated characteristics.

Column-I **Column-II**

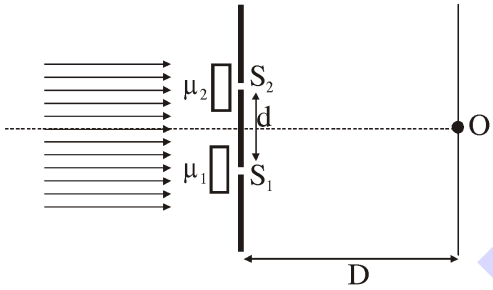
(A)



monochromatic point source S placed in focal plane.

(P) Zero order maxima lies above O.

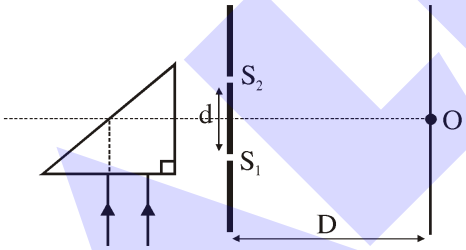
(B)



monochromatic parallel beam incident on S_1S_2 through transparent slabs of same thickness but $\mu_1 > \mu_2$

(Q) If a transparent mica sheet is introduced infront of S_2 central bright fringe can be obtained at O.

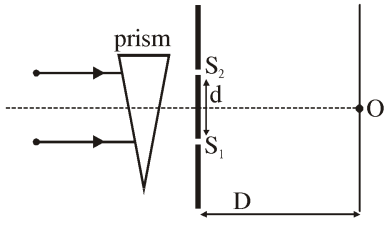
(C)



monochromatic parallel beam incident on a right angled isosceles prism of refractive index 1.50

(R) Fringe width $\beta = \frac{\lambda D}{d}$

(D)



monochromatic parallel beam incident on thin prism

(S) Point O can be a minima

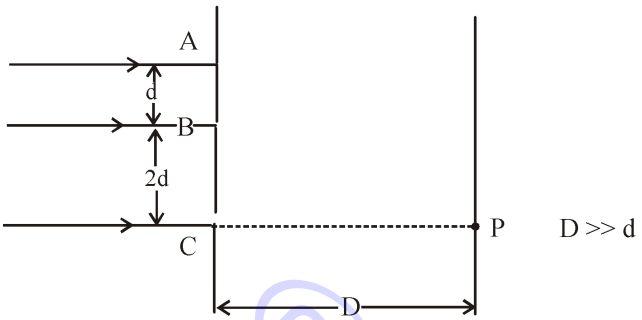
(T) Point O can be a least order minima.

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23. In the figure A, B and C are three slits. Intensity of light from each slit is I_0 at point P. Wavelength of light is λ. Match the conditions of column-I to resultant intensity at point P given in column-II.

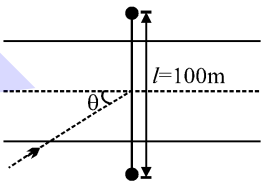
Column - I	Column-II
(A) If $\frac{d^2}{D}=\lambda$	(P) $I_p = I_o$
(B) If $\frac{d^2}{D}=\frac{\lambda}{2}$	(Q) $I_p = 4I_o$
(C) If $\frac{d^2}{D}=\frac{\lambda}{4}$	(R) $I_p = 5I_o$
(D) If $\frac{d^2}{D}=2\lambda$	(S) $I_p = 9I_o$
	(T) $I_p = 3I_o$



Subjective Type

5 Q. [4 M (0)]

24. Two radio transmitter each transmitting a radio of frequency 12 MHz in phase with each other are placed at a distance l = 100 m on either side of the runway on the towers as shown in the figure. An aircraft with a ground speed of V is flying towards the airport such that its velocity makes an angle θ with the runway. The aircraft is heading directly for the mid-point of the towers.

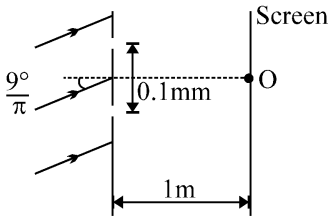


(i) At the aircraft's current position, the intensity of the signal from each tower separately would be I_0. Find the intensity of the combined signal from both towers received by the aircraft for heading of θ = 0 and θ = π/2.

(ii) For what heading would the aircraft receive no signal. Assuming distance of plane much greater than l.

(iii) If the aircraft is heading from large distance directly towards one of the transmitter in the direction perpendicular to 'l' find the position of the aircraft from the nearest tower when the signal dropped to a minimum first time.

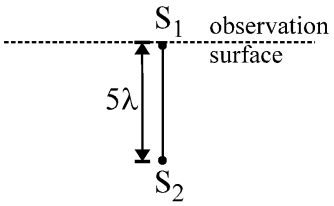
25. In a Young's double slit experiment, a parallel beam of light of wavelength 6000 Å is incident on the slit at an angle of $\frac{9°}{\pi}$ as shown in the figure.



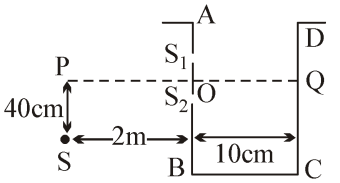
(i) Find fringe width of the interference pattern.

(ii) Find phase difference between the light coming from the two slits at O.

26. Two microwave coherent point sources emitting waves of wavelength λ are placed at 5λ distance apart. The interference is being observed on a flat non-reflecting surface along a line passing through one source, in a direction perpendicular to the line joining the two sources (refer figure). Considering λ as 4 mm, calculate the positions of maxima and draw shape of interference pattern. Take initial phase difference between the two sources to be zero.



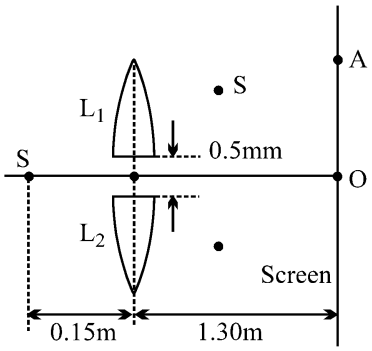
27. A vessel *ABCD* of 10 cm width has two small slits S_1 and S_2 sealed with identical glass plates of equal thickness. The distance between the slits is 0.8 mm. *POQ* is the line perpendicular to the plane *AB* and passing through *O*, the middle point of S_1 and S_2. A monochromatic light source is kept at *S*, 40cm below *P* and 2 m from the vessel, to illuminate the slits as shown in the figure below. Calculate the position of the central bright fringe on the other wall *CD* with respect to the line *OQ*. Now, a liquid is poured into the vessel and filled up to *OQ*. The central bright fringe is found to be at *Q*. Calculate the refractive index of the liquid.



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28. In the figure shown S is a monochromatic point source emitting light of wavelength $\lambda = 500$ nm. A thin lens of circular shape and focal length 0.10 m is cut into two identical halves L_1 and L_2 by a plane passing through a diameter. The two halves are placed symmetrically about the central axis SO with a gap of 0.5 mm. The distance along the axis from S to L_1 and L_2 is 0.15 m, while that from L_1 & L_2 to O is 1.30 m. The screen at O is normal to SO.



(i) If the third intensity maximum occurs at the point A on the screen, find the distance OA.

(ii) If the gap between L_1 & L_2 is reduced from its original value of 0.5 mm, will the distance OA increase, decrease or remain the same ?

GR # WAVE OPTICS **ANSWER KEY**

SECTION-I

Single Correct Answer Type **8 Q. [3 M (–1)]**

1. Ans. (A) **2. Ans. (A)** **3. Ans. (D)** **4. Ans. (A)** **5. Ans. (B)**
6. Ans. (A) **7. Ans. (C)** **8. Ans. (A)**

Multiple Correct Answer Type **8 Q. [4 M (–1)]**

9. Ans. (A,B,D) **10. Ans. (B,D)** **11. Ans. (A,D)** **12. Ans. (B, C)** **13. Ans. (A,B,C)**
14. Ans. (A,B,C) **15. Ans. (B,C,D)** **16. Ans. (A,B)**

SECTION-II

Numerical Answer Type Question (upto second decimal place) **3Q.[3M (0)]**

17. Ans. 9.3 μm **18. Ans. 1.25** **19. Ans.** 0.2 mm

SECTION-III

Numerical Grid Type (Ranging from 0 to 9) **1 Q. [4 M (0)]**

20. Ans. 5

SECTION-IV

Matrix Match Type (4 × 5) **3Q. [8 M (for each entry +2(0)]**

21. Ans. (A) → (R) ; (B) → (T) ; (C) → (Q,R) ; (D) → (P,S)
22. Ans. (A) → (P,R,S,T) ; (B) → (Q,R,S,T),; (C) → (R) ; (D) → (P,R,S,T)
23. Ans. (A) → (P) ; (B) → (R) ; (C) → (P) ; (D) → (S)

Subjective Type **5 Q. [4 M (0)]**

24. Ans. $4I_0$, $\theta = \sin^{-1}\left(\frac{2n+1}{8}\right)$ n = 0, 1, 2, 3, (iii) 393.75 m **25. Ans.** (i) 6 mm, (ii) $50\pi/3$

26. Ans. 48, 21, $\frac{32}{3}$, $\frac{9}{2}$, 0 m.m.; 

27. Ans. (i) y = 2 cm, (ii) m = 1.0016

28. Ans. (i) 1 mm (ii) increase

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